A

PBL Project Report Entitled On

SMART SENTIMENT ANALYZER USING MACHINE LEARNING

LEARNING

In partial fulfilment of the requirements for the award of

BACHELORS OF TECHNOLOGY

In

Computer Science Engineering(AI & ML)

Submitted by

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HYDERABAD INSTITUTE OF TECHNOLOGY AND MANAGEMENT

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DECLARATION

We, Inderjeet Karan Jaiswal (24E51A6665), Akash Bhuyan (24E51A6608), and Dixit Devra (24E51A6649), students of Bachelor of Technology in CSM, session 2025–2026, Hyderabad Institute of Technology and Management, Gowdavelly, Hyderabad, Telangana State, hereby declare that the work presented in this project entitled "**SMART SENTIMENT ANALYZER USING MACHINE LEARNING**" is the outcome of our own bona fide effort and is correct to the best of our knowledge. This work has been carried out adhering to engineering ethics. It contains no material previously published or written by any other person, nor material accepted for the award of any other degree or diploma of the university or any other institute of higher learning, except where due acknowledgement has been made in the text.

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TABLE OF CONTENTS

LIST OF FIGURES

ABSTRACT

Sentiment analysis is an important technique used to understand customer opinions from
product reviews. It helps organizations know whether customers feel positive, negative, or neutral about a product. In this project, a machine learning–based system is developed to automatically classify customer sentiment using textual product reviews.

The project uses both traditional machine learning algorithms and modern deep learning
models for sentiment classification. Traditional algorithms such as Support Vector Machine (SVM) and Decision Tree are implemented using text feature extraction methods like Bag-of-Words and TF-IDF. These methods convert text into numerical form so that machine learning models can process it. Traditional models are simple, fast, and easy to implement, but they struggle to understand the context and meaning of words in longer sentences.

To overcome these limitations, deep learning models such as Recurrent Neural Networks (RNN) and Transformer-based models are used. RNN models analyze text in a sequence, which helps in understanding word order and sentence structure. Transformer models use attention mechanisms to focus on important words in a review, allowing them to capture deeper contextual meaning and relationships between words more effectively.

All models are evaluated and compared using performance metrics such as accuracy, precision, and recall. The results show that deep learning models provide better sentiment classification performance than traditional machine learning algorithms. However, traditional models require fewer computational resources and work well for smaller datasets. This project highlights the strengths and limitations of different approaches and helps in selecting the appropriate model for customer sentiment analysis.

1. INTRODUCTION

1.1 Background & Context of the Project

In today's digital world, customers frequently share their opinions and experiences through online product reviews, social media, and feedback platforms. These reviews play a crucial role in influencing purchasing decisions and shaping the reputation of businesses. However, analyzing a large number of customer reviews manually is time-consuming, inefficient, and prone to errors.

With the rapid growth of e-commerce platforms, the volume of customer feedback has increased significantly. Businesses require automated systems to analyze this data and extract meaningful insights. Sentiment Analysis, a subfield of Natural Language Processing (NLP), helps in identifying whether a piece of text expresses a positive, negative, or neutral opinion.

This project focuses on developing a Customer Sentiment Analysis system using Machine Learning and Deep Learning techniques. The system processes textual data from product reviews and automatically classifies the sentiment, enabling organizations to make data-driven decisions and improve customer satisfaction.

1.2 Justification for Project Selection

This project was chosen because understanding customer sentiment is essential for businesses to improve their products and services. Manual analysis of reviews is not practical when dealing with large datasets, and it often leads to delayed decision-making.

Traditional approaches fail to capture the complexity and context of human language, especially in cases involving sarcasm, mixed opinions, or informal text. Therefore, there is a need for intelligent systems that can automatically analyze and interpret customer feedback accurately.

By implementing both traditional Machine Learning models and advanced Deep Learning models, this project provides a comparative study of different approaches. It helps in identifying the most effective method for sentiment classification while balancing accuracy and computational efficiency.

1.3 Problem Statement

Organizations face significant challenges in analyzing large volumes of customer reviews efficiently and accurately. Manual processing is slow and not scalable, while simple automated systems often fail to understand the context and nuances of language.

Common issues include:

Difficulty in handling large datasets

Inability to capture contextual meaning

Misclassification due to ambiguous or complex sentences

Lack of real-time analysis

Therefore, there is a need for an automated sentiment analysis system that can process large amounts of textual data, accurately classify sentiments, and provide meaningful insights. The system should also compare different machine learning approaches to determine the most effective solution.

1.4 Objectives of the Project

The main objective of this project is to develop an efficient system for automatic sentiment classification of customer reviews using Machine Learning and Deep Learning techniques.

The specific objectives include:

1: Designing a system to classify product reviews into positive, negative, and neutral sentiments.

2: Implementing traditional Machine Learning models such as Support Vector Machine (SVM) and Decision Tree.

3: Applying Deep Learning models such as Recurrent Neural Networks (RNN) and Transformer-based models.

4: Performing text preprocessing techniques like tokenization, stopword removal, and normalization.

5: Converting textual data into numerical form using techniques like Bag-of-Words and TF-IDF.

6: Comparing the performance of different models based on accuracy, precision, and recall.

7: Identifying the most efficient model for real-world sentiment analysis applications.

Overall, this project aims to provide a reliable and scalable solution for analyzing customer sentiment and supporting better business decision-making.

2. LITERATURE REVIEW / RELATED WORK

Sentiment Analysis has become an important research area in the fields of Natural Language Processing (NLP) and Machine Learning. It focuses on identifying and classifying opinions expressed in text into categories such as positive, negative, or neutral. With the rapid growth of online platforms, a large amount of user-generated content is available, making automated sentiment analysis essential for businesses and researchers.

Early approaches to sentiment analysis were based on rule-based and lexicon-based methods. These methods use predefined dictionaries of words with associated sentiment scores. While they are simple and easy to implement, they often fail to handle complex language patterns such as sarcasm, negation, and context-dependent meanings.

Traditional Machine Learning techniques such as Support Vector Machine (SVM), Naive Bayes, and Decision Trees have been widely used for sentiment classification. These models rely on feature extraction techniques like Bag-of-Words and TF-IDF to convert text into numerical form. They provide good performance for structured datasets but have limitations in understanding the sequence and context of words in a sentence.

With advancements in deep learning, models such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks have been introduced. These models are capable of capturing sequential dependencies in text data, allowing better understanding of sentence structure and context. However, they may require more training time and computational resources.

More recently, Transformer-based models such as BERT (Bidirectional Encoder Representations from Transformers) have gained popularity in sentiment analysis tasks. These models use attention mechanisms to focus on important words in a sentence and capture contextual relationships more effectively than traditional models. Transformers have achieved state-of-the-art performance in many NLP tasks, including sentiment classification.

Several tools and applications have been developed for sentiment analysis, including social media monitoring systems and customer feedback analysis platforms. However, many existing systems either rely on traditional models with limited accuracy or require high computational power when using advanced deep learning models.

From the literature, it is observed that while traditional Machine Learning models are efficient and easy to implement, they lack the ability to capture deep contextual meaning. On the other hand, deep learning models provide higher accuracy but require more resources. Therefore, there is a need to compare these approaches and identify the most suitable method based on performance and practical constraints.

3. METHODOLOGY

The methodology used for developing the Smart Sentiment Analyzer using Machine Learning consists of several structured phases including data collection, preprocessing, feature extraction, model training, evaluation, and deployment. The system is designed to automatically classify customer reviews into positive, negative, and neutral sentiments using both traditional Machine Learning algorithms and advanced Deep Learning models.

3.1 Data Sources & Collection

The project begins with collecting customer review datasets from publicly available sources and online review platforms. The collected dataset contains textual reviews along with sentiment labels.

The data includes:

- Product reviews from e-commerce platforms
- Customer feedback comments
- Social media text samples
- Sentiment labels such as Positive, Negative, and Neutral
- Review metadata including ratings and timestamps

The dataset used for training and testing contains thousands of customer reviews, which provide sufficient diversity for sentiment classification.

3.2 Data Preprocessing & Cleaning

Before training the models, the collected textual data is preprocessed and cleaned to improve accuracy and remove noise.

3.2.1 Identified Data Issues

During data inspection, several issues were identified:

- Duplicate customer reviews

- Missing or incomplete text entries
- Presence of special characters and emojis
- Spelling mistakes and inconsistent formatting
- Stop words that do not contribute to sentiment
- Unnecessary punctuation and hyperlinks
- Imbalanced sentiment classes

3.2.2 Data Cleaning Steps

To prepare the dataset for sentiment analysis, the following preprocessing steps were applied:

1. Duplicate reviews were removed.
2. Null and empty values were handled.
3. Text was converted into lowercase format.
4. Stop words such as "is", "the", and "and" were removed.
5. Tokenization was performed to split text into individual words.
6. Lemmatization and stemming techniques were used for normalization.
7. URLs, hashtags, special symbols, and punctuation marks were removed.
8. Text data was transformed into numerical vectors using Bag-of-Words and TF-IDF techniques.

These preprocessing techniques helped improve model performance and reduced unnecessary computational complexity.

3.3 Tools and Technologies Used

The project uses various Machine Learning, Deep Learning, and Web technologies for implementation.

Programming Language

- Python

Libraries and Frameworks

- Scikit-learn – Machine Learning model implementation
- TensorFlow / Keras – Deep Learning model development
- Pandas – Data handling and preprocessing
- NumPy – Numerical computations
- NLTK – Natural Language Processing tasks
- Matplotlib and Seaborn – Data visualization

Web Technologies

- HTML – Structure of the web application
- CSS – Styling and responsive design
- JavaScript – Interactive frontend functionality
- Flask – Backend integration and API handling

Development Tools

- Jupyter Notebook – Model experimentation
- VS Code – Development environment
- GitHub – Version control and project management

These technologies provide an efficient and scalable environment for sentiment analysis.

3.4 KPI Selection Rationale

Key Performance Indicators (KPIs) were selected to evaluate the effectiveness of the sentiment analysis system.

The major KPIs include:

- Accuracy – Measures the overall correctness of predictions
- Precision – Measures how many predicted sentiments are correct
- Recall – Measures the model's ability to identify actual sentiments

- F1-Score – Balances precision and recall
- Training Time – Measures computational efficiency
- Prediction Speed – Determines real-time usability

These KPIs help compare the performance of traditional Machine Learning models and Deep Learning approaches.

3.5 Final Data State and Analysis Readiness

After preprocessing, the final dataset becomes clean, normalized, and suitable for model training.

The processed dataset includes:

- Tokenized review text
- Numerical feature vectors
- Balanced sentiment classes
- Training and testing datasets

The dataset is then divided into:

- Training Data – Used to train the models
- Testing Data – Used to evaluate model performance

This prepared data ensures efficient training and accurate sentiment prediction.

3.6 Model Development and Training

Multiple Machine Learning and Deep Learning models were implemented and compared.

Traditional Machine Learning Models

1. Support Vector Machine (SVM)
2. Decision Tree
3. Naive Bayes

These models were trained using TF-IDF and Bag-of-Words features.

Deep Learning Models

1. Recurrent Neural Network (RNN)
2. Long Short-Term Memory (LSTM)
3. Transformer-based Models

Deep Learning models were trained using word embeddings and sequential text representations to better capture contextual meaning.

Sample Code

Machine Learning Model (Python)

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

reviews = [
"This product is amazing",
"Worst experience ever",
"Good quality and fast delivery",
"Not satisfied with the product"
]

labels = [1, 0, 1, 0]
```

```
vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(reviews)

X_train, X_test, y_train, y_test = train_test_split(
X, labels, test_size=0.2, random_state=42
)

model = SVC()
model.fit(X_train, y_train)

predictions = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, predictions))
```

Frontend Interface (HTML)

```
<html>
<head>
<title>Smart Sentiment Analyzer</title>
</head>
<body>
<h1>Sentiment Analysis System</h1>
<textarea placeholder="Enter your review"></textarea>
<button>Analyze Sentiment</button>
</body>
</html>
```

4. DATA ANALYSIS

4.1 Key Performance Indicators (KPIs)

The following KPIs were used to measure the efficiency and performance of the sentiment analysis system:

- Model Accuracy
- Precision and Recall
- F1-Score
- Training Time
- Prediction Speed
- Classification Consistency

These indicators help determine the reliability and effectiveness of the developed models.

4.2 Descriptive Analysis

Descriptive analysis was performed to understand the structure and distribution of the dataset.

The analysis includes:

- Total number of reviews analyzed
- Distribution of positive, negative, and neutral sentiments
- Average review length
- Frequently occurring words
- Word frequency visualization using word clouds and bar graphs

This analysis helped identify patterns in customer opinions and overall dataset characteristics.

4.3 Comparative Analysis

Comparative analysis was performed to evaluate the performance of different models. The comparison includes:

- Traditional Machine Learning vs Deep Learning models
- SVM vs Decision Tree performance
- RNN vs Transformer-based models
- Accuracy and computational efficiency comparison

The results showed that:

- Traditional models are faster and require fewer resources.
- Deep Learning models provide higher accuracy and better contextual understanding.
- Transformer models achieved the best sentiment classification performance.

4.4 Data Mining Techniques

Several data mining and NLP techniques were used in the project:

- Classification – Categorizing reviews into sentiment classes
- Feature Extraction – TF-IDF and Bag-of-Words representation
- Text Mining – Extracting meaningful patterns from reviews
- Clustering – Grouping reviews with similar sentiments
- Word Embeddings – Capturing semantic meaning of words

These techniques helped improve the quality and accuracy of sentiment prediction.

5. RESULTS & INTERPRETATION

5.1 Key Findings

The following major findings were observed during project implementation:

- Deep Learning models achieved higher accuracy than traditional Machine Learning models.
- Transformer-based models provided the best contextual understanding.
- Traditional models performed well on smaller datasets with lower computational cost.
- Proper preprocessing significantly improved prediction accuracy.
- TF-IDF feature extraction improved the performance of traditional models.

5.2 Results

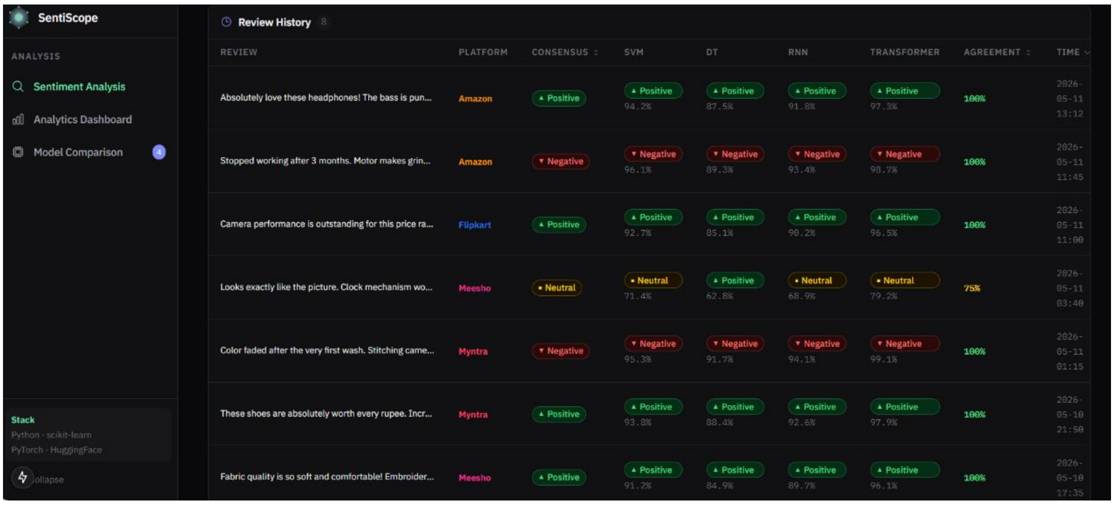


Figure 6.2.1 Review History

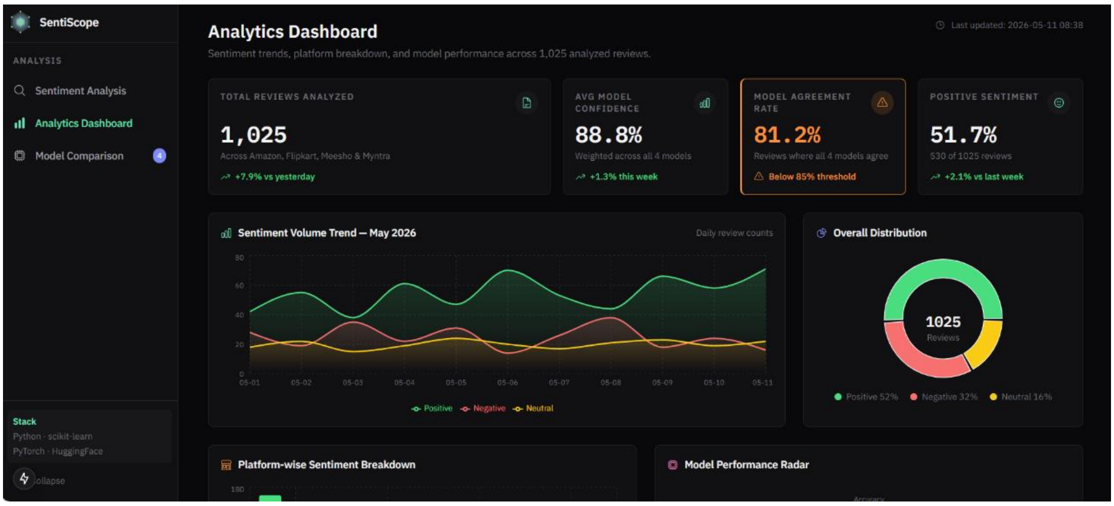


Figure 6.2.2 Accuracy of the model

Figure 6.2.3 Platform wise Sentiment Breakdown

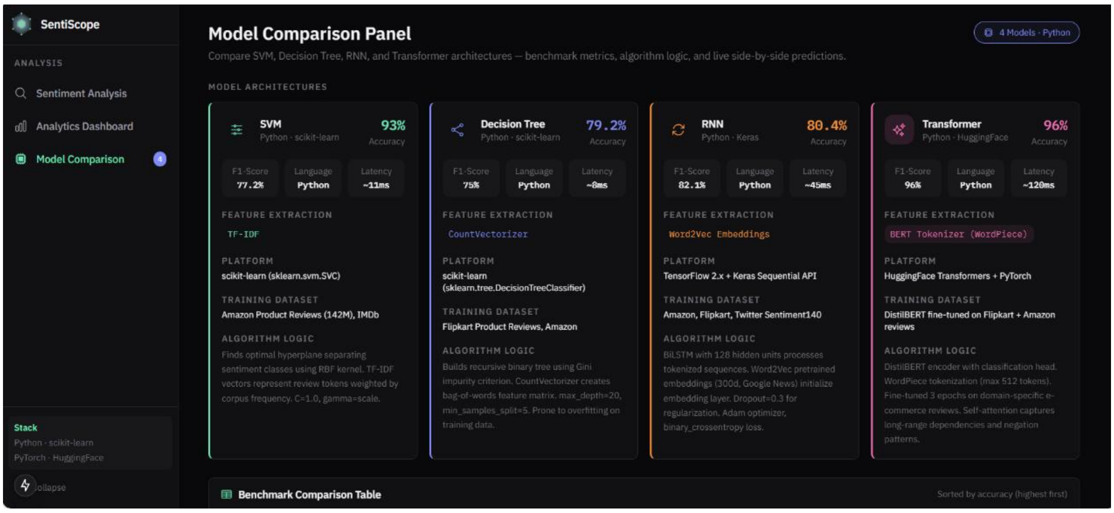
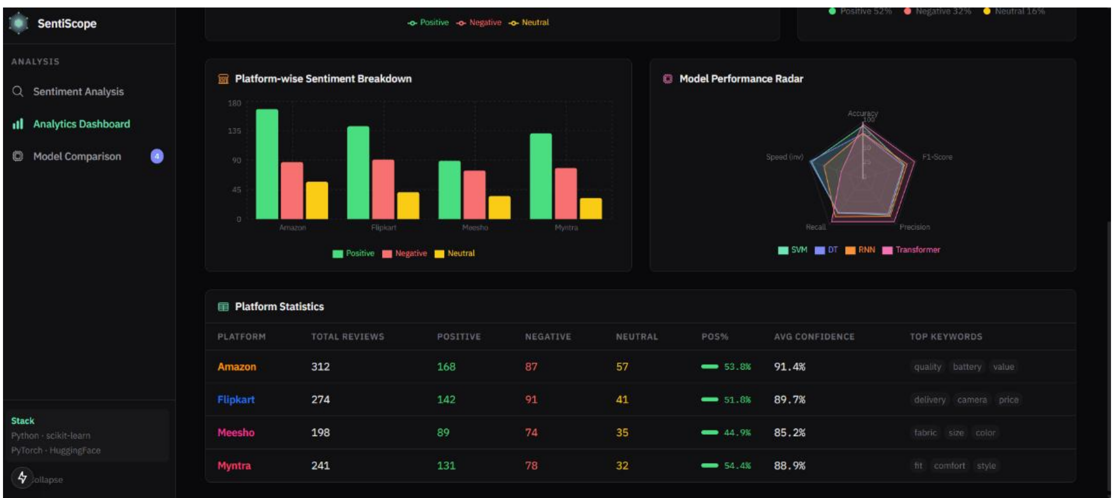


Figure 6.2.4 Comparison of models



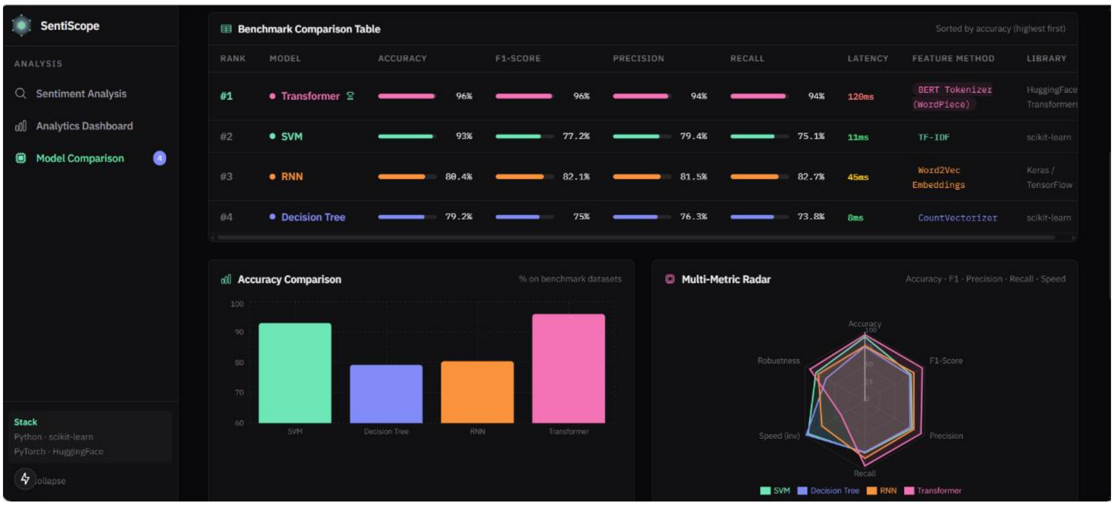


Figure 6.2.5 Benchmark Comparison Table

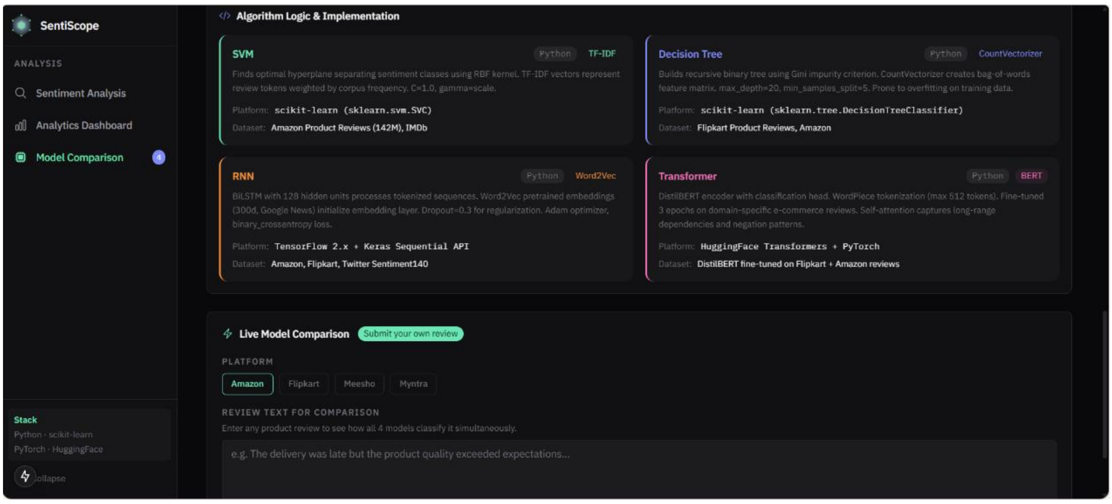


Figure 6.2.6 Algorithm logic and Implementation

The sentiment analysis system successfully classified customer reviews into positive, negative, and neutral categories.
The final results demonstrated:

- Accurate sentiment prediction for customer reviews
- Better contextual analysis using Deep Learning techniques
- Faster prediction using traditional Machine Learning models
- Improved visualization of customer opinion trends

The project successfully compared multiple approaches and identified the strengths and limitations of each technique.

6. DISCUSSION

The development of the Smart Sentiment Analyzer highlighted several technical challenges and important learning experiences.

One of the major challenges was handling unstructured textual data. Customer reviews often contain spelling mistakes, abbreviations, emojis, sarcasm, and informal language, making sentiment prediction difficult. Proper preprocessing and normalization techniques were essential to improve model performance.
Another challenge was balancing computational efficiency with prediction accuracy. Traditional Machine Learning models required less training time and computational power but struggled with contextual understanding. On the other hand, Deep Learning models such as RNNs and Transformers achieved higher accuracy but required more processing power and larger datasets.
The project also faced challenges related to dataset imbalance. Some sentiment classes contained significantly more samples than others, which affected prediction fairness. Data balancing and preprocessing methods helped reduce this issue.
Additionally, selecting suitable hyperparameters for Deep Learning models required repeated experimentation. Different configurations produced varying results, making model optimization an important part of the project.

6.1 Challenges & Limitations

- Difficulty in understanding sarcasm and complex language
- Requirement of large datasets for Deep Learning models
- High computational cost for Transformer-based models
- Handling noisy and unstructured text data
- Model accuracy depends heavily on preprocessing quality

6.2 Lessons Learned

Several important lessons were learned during the development of this project:

- Data preprocessing plays a critical role in NLP applications.
- Feature extraction techniques significantly affect model performance.
- Deep Learning models provide better contextual understanding.
- Model comparison is important for selecting suitable algorithms.
- Proper dataset balancing improves prediction reliability.

Overall, the project provided practical knowledge about Machine Learning, Deep Learning, and Natural Language Processing techniques.

7. CONCLUSION

The Smart Sentiment Analyzer using Machine Learning successfully demonstrates how Artificial Intelligence can be used to automatically analyze customer opinions from textual reviews. The system effectively classifies customer feedback into positive, negative, and neutral sentiments using both traditional Machine Learning and advanced Deep Learning approaches.
Traditional Machine Learning models such as Support Vector Machine and Decision Tree provided fast and computationally efficient solutions for sentiment classification. However, they faced limitations in understanding the contextual meaning of complex sentences.
Deep Learning models such as Recurrent Neural Networks and Transformer-based architectures achieved significantly better performance by understanding sequential and contextual relationships between words. Among all models, Transformer-based approaches produced the highest accuracy and best sentiment understanding.
The project also highlights the importance of preprocessing techniques such as tokenization, stopword removal, normalization, and feature extraction. These steps greatly improved model performance and prediction quality.
This project provides a scalable and practical solution for businesses to analyze customer feedback efficiently. It reduces manual effort, improves decision-making, and helps organizations better understand customer opinions.

In conclusion, the Smart Sentiment Analyzer serves as an effective application of Machine Learning and Deep Learning in Natural Language Processing and demonstrates the growing importance of AI-driven customer feedback analysis systems.

8. RECOMMENDATIONS

Real-Time Sentiment Analysis

Future versions of the system can support real-time sentiment prediction for live customer reviews and social media comments.

Multilingual Support

The system can be enhanced to analyze reviews written in multiple languages.

Improved Deep Learning Models

Advanced Transformer architectures such as BERT and GPT-based models can further improve contextual understanding and accuracy.

Integration with Business Platforms

The system can be integrated with e-commerce websites, CRM software, and social media platforms for automated customer feedback analysis.

Visualization Dashboards

Interactive dashboards and analytics charts can be added to help businesses monitor customer satisfaction trends.

9. REFERENCES

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